power function (704) a function in the form $y = x^b$, where b is a real number

prediction equation (38) an equation suggested by the points of a scatter plot used to predict other points

pre-image (88) the graph of an object before a transformation

principal values (406, 456) the unique solutions of a trigonometric equation if the values of the function are restricted to two adjacent quadrants

prisoner point (816) If the iterates do not approach infinity for some initial value, that point is called a prisoner point.

prisoner set (599) the set of initial values for which the iterates of a function do not approach infinity

probability (852) the measure of the chance of a desired outcome happening

pure imaginary number (206, 581) the complex number $a + b\boldsymbol{i}$ when $a = 0$ and $b \neq 0$

Q

quadrantal angle (278) an angle in standard position whose terminal side coincides with one of the axes

quartile (909) one of four groupings of a set of data determined by the median of the set and the medians of the sets determined by the median

R

radian (343) the measure of a central angle whose sides intercept an arc that is the same length as the radius of the circle

radical equation (251) an equation that contains a radical expression with the variable in the radicand

radical inequality (253) an inequality that contains a radical expression with the variable in the radicand

radius See circle.

random sample (927) a sample in which every member of the population has an equal chance to be selected

range (890) the difference of the greatest and least values in a set of data

range of a relation (5) the set of all ordinates of the ordered pairs of a relation

rate of change (956) the derivative of a function when applied to real-world applications

rational equation (243) an equation that consists of one or more rational expressions

rational function (180) the quotient of two polynomials in the form $f(x) = \frac{g(x)}{h(x)}$, where $h(x) \neq 0$

rational inequality (245) an inequality that consists of one or more rational expressions

ratio test (785) a method to test convergence in an infinite series

real part of a complex number (581) a in the complex number $a + b\boldsymbol{i}$

rectangular form (581) a complex number written as $x + y\boldsymbol{i}$, where x is the real part and $y\boldsymbol{i}$ is the imaginary part

rectangular hyperbola (648) A special case of the equilateral hyperbola, where the coordinate axes are the asymptotes. The general equation of a rectangular hyperbola is $xy = c$, where c is a nonzero constant.

recursive formula (758) a formula used for determining the next term of a sequence using one or more of the previous terms

reduced sample space (862) the subset of a sample space that contains only those outcomes that satisfy a given condition

reduction identity (446) identity that involves adding and subtracting the quandrantal angles, 90°, 180°, and 270°, from the angle measure to find equivalent values of a trigonometric function

reference angle (280) the acute angle formed by the terminal side of an angle in standard position and the x-axis

reflection (88) a linear transformation that flips a figure over a line called the line of symmetry

reflection matrix (89) a matrix used to reflect an object over a line or plane

regression line (40) a best-fit line

relation (5) a set of ordered pairs

relative extremum (172) a point that represents the maximum or minimum for a certain interval

relative maximum (172) a point that represents the maximum for a certain interval

relative minimum (172) a point that represents the minimum for a certain interval

resultant of vectors (486) the sum of two or more vectors

root (206) a solution of the equation $P(x) = 0$

rose (563) The graph of a polar equation of the form $r = a \cos n\theta$ or $r = a \sin n\theta$.

rotation (88) a transformation in which an object is moved around a center point

rotation matrix (91) a matrix used to rotate an object

row matrix (78) a matrix that has only one row

S

sample space (852) the set of all possible outcomes of an event

CONTENTS